

Backend Engineering Fundamentals - 6 Week Intensive Course

Week 4: Authentication, Authorization & Security

Learning Objectives

- Implement user registration and login
- Understand JWT tokens and session management
- Learn password hashing and security best practices
- Implement role-based access control
- Secure your API endpoints

Resources

Videos:

- [User Authentication in Flask](#) by Tech With Tim (30 mins)
- [JWT Authentication Tutorial](#) by CalebTheVideoMaker2 (30 mins)

Text Resources:

- [Flask-JWT-Extended Documentation](#)
- [OWASP Top 10 Security Risks](#)
- [Real Python - Token-Based Authentication](#)

Topics Covered

- Password hashing (bcrypt)
- User registration and login flows
- JWT tokens (access and refresh tokens)
- Protected routes and decorators
- Role-based access control (RBAC)
- CORS (Cross-Origin Resource Sharing)

- Input sanitization
- SQL injection prevention
- Environment variables for secrets
- Rate limiting basics

Week 4 Project: Multi-User Blog API

Build a blog platform with authentication

Requirements:

- User registration (username, email, password)
- User login (return JWT token)
- Protected endpoints (require authentication)
- Create blog posts (authenticated users only)
- Update/delete own posts only
- Public endpoint to view all posts
- User roles (admin can delete any post)
- Password requirements validation
- Token refresh endpoint
- User profile endpoint

This project tests: Authentication, authorization, JWT implementation, security practices, and access control.